# Money Plot Labs Technical Whitepaper
# The Pure Math of Financial Freedom

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**Abstract**

This whitepaper derives the explicit, closed-form analytical solution for a fixed-horizon retirement timeline. By presenting a simple linear model first, we establish a transparent baseline using middle-school algebra. We then expand this model into exponential growth curves by mapping the exact continuous boundaries between an asset accumulation phase and an annuity-due depletion phase. This isolates the exact number of required working years ($w$) as a function of savings rate, spending targets, initial capital, investment growth, and desired legacy floors.

## 1 Introduction & Variables

Standard personal finance paradigms rely on fragile rules of thumb. To achieve absolute mathematical certainty, we model an individual's financial lifecycle across a strictly bounded timeline from a starting age ($T_{\text{start}}$) to a maximum life expectancy ($T_{\text{end}}$). We define the total planning horizon as:

$$H = T_{\text{end}} - T_{\text{start}} \tag{1}$$

Within this horizon $H$, an individual works for $w$ consecutive years and spends the remaining $f$ years in full retirement, such that:

$$H = w + f \tag{2}$$

### 1.1 Parameter Definitions

- $A_0$: Initial Net Worth / Starting Balance ($)
- $c$: Annual Cash Savings Flow during working years ($)
- $b$: Annual Retirement Budget Consumption ($)
- $L$: Desired Capital Legacy Floor / Inheritance Remaining at Death ($)
- $r$: Real Growth Rate adjusted for inflation
- $w$: Accumulation Phase duration (Working Years)
- $f$: Depletion Phase duration (Retirement Years)

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# 2 The Linear Baseline Case ($r = 0\%$)

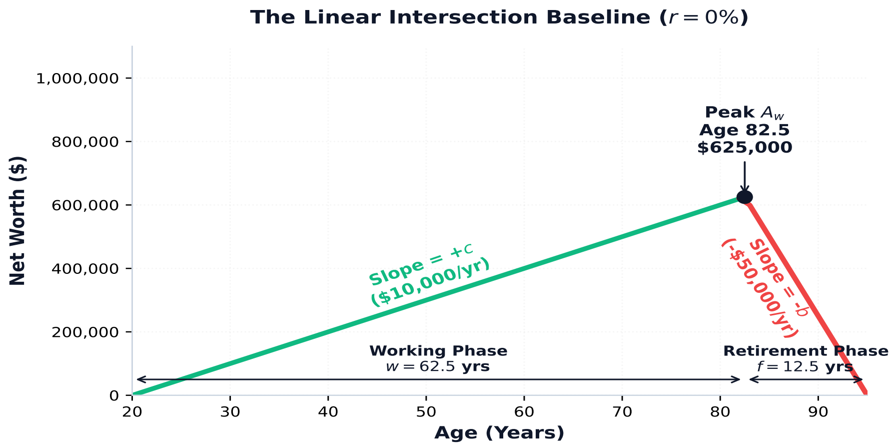


Figure 1: The straight-line boundary crossover mapping where $r = 0\%$. Assets accumulate smoothly via labor and deplete symmetrically via consumption until hitting the legacy boundary.

Before introducing the compounding curves of financial markets, we establish an intuitive baseline by assuming real asset growth perfectly tracks inflation ($r = 0\%$). In this environment, wealth accumulates and depletes along perfectly straight lines, making it easy to see how the timelines cross. For simplicity we also assume a zero starting balance ($A_0 = 0$) and no remaining inheritance ($L = 0$).

## 2.1 The First-Principles Equations

During the working phase, the accumulation of assets grows linearly from our starting balance $A_0$ by injecting our cash savings flow $c$ at the end of each year for $w$ years:

$$\text{Peak Assets} = c \cdot w \tag{3}$$

Upon entering retirement, the accumulation engine shuts off and the depletion engine takes over. The capital depletion loop draws down our peak nest egg by a flat budget spending $b$ each year for the remaining $f$ retirement years $(H - w)$ until landing exactly on our target legacy floor $L$:

$$\text{Peak Assets Required} = b \cdot f \tag{4}$$

## 2.2 The Closed-Form Linear Solution

To find the precise year where working ends and retirement begins, we set the asset accumulation equation equal to the asset depletion requirement:

$$c \cdot w = b \cdot f \tag{5}$$

Substitute $f$ with $(H - w)$ and distributing the retirement consumption $b$ across the timeline parameters yields:

$$c \cdot w = b \cdot H - b \cdot w \tag{6}$$

Grouping all terms containing our target variable $w$ onto the left-hand side:

$$c \cdot w + b \cdot w = b \cdot H \tag{7}$$

Factoring out $w$ isolates our terms neatly:

$$w(c + b) = b \cdot H \tag{8}$$

Dividing through by $(c + b)$ yields the beautiful, closed-form **Money Plot Linear Baseline Equation**:

$$w = \frac{b \cdot H}{c + b} \tag{9}$$

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# 3 Phase 1: The Exponential Accumulation Engine

When we introduce real market investment returns ($r > 0\%$), our straight lines transform into curves. During the working phase ($w$), net worth grows via compound interest on the initial principal while absorbing an ordinary annuity flow of annual savings $c$ injected at the end of each period. The net worth at the exact boundary of retirement ($A_w$) is defined by:

$$A_w = A_0(1 + r)^w + c \cdot \frac{(1 + r)^w - 1}{r} \tag{10}$$

## 3.1 Step-by-Step Inductive Derivation of $A_w$

To understand the structural origin of the accumulation equation, we trace the boundary state changes sequentially at the close of each annual compounding cycle:

$$\begin{aligned}
A_1 &= A_0(1 + r) + c \\
A_2 &= A_1(1 + r) + c = [A_0(1 + r) + c]\,(1 + r) + c \\
&= A_0(1 + r)^2 + c(1 + r) + c \\
A_3 &= A_2(1 + r) + c = \left[A_0(1 + r)^2 + c(1 + r) + c\right](1 + r) + c \\
&= A_0(1 + r)^3 + c(1 + r)^2 + c(1 + r) + c
\end{aligned}$$

Extending this inductive sequence to a generalized working timeline of $w$ years yields:

$$A_w = A_0(1 + r)^w + c \sum_{k=0}^{w-1} (1 + r)^k \tag{11}$$

The summation block represents a classic geometric progression. Applying the standard Geometric Series Summation Theorem where the common ratio is $(1 + r)$, the series condenses analytically:

$$\sum_{k=0}^{w-1} (1 + r)^k = \frac{(1 + r)^w - 1}{(1 + r) - 1} = \frac{(1 + r)^w - 1}{r} \tag{12}$$

Substituting Equation 12 back into Equation 11 yields our verified final accumulation master expression matching Equation 10.

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# 4 Phase 2: The Exponential Depletion Engine

Upon entering retirement, the cash flow behavior flips. To match real-world physical constraints, living expenses must be accessible at the *beginning* of each calendar year. Therefore, retirement spending is modeled strictly as an **Annuity Due**.

$$R_f = A_w(1+r)^f - b \cdot \frac{(1+r)^{f+1} - (1+r)}{r} \tag{13}$$

## 4.1 Step-by-Step Inductive Derivation of the Depletion Phase

We trace the remaining capital balance $R$ at the close of each retirement year sequentially, demonstrating why the timing mismatch injects an extra factor of $(1+r)$ compared to an ordinary annuity:

$$\begin{aligned}
R_1 &= A_{w+1} \\
R_1 &= (A_w - b)(1+r) = A_w(1+r) - b(1+r) \\
R_2 &= (R_1 - b)(1+r) = [A_w(1+r) - b(1+r) - b]\,(1+r) \\
&= A_w(1+r)^2 - b(1+r)^2 - b(1+r) \\
R_3 &= (R_2 - b)(1+r) = \left[A_w(1+r)^2 - b(1+r)^2 - b(1+r) - b\right](1+r) \\
&= A_w(1+r)^3 - b(1+r)^3 - b(1+r)^2 - b(1+r)
\end{aligned}$$

Extending this progression to a total retirement duration of $f$ years yields:

$$R_f = A_w(1+r)^f - b\sum_{k=1}^{f}(1+r)^k \tag{14}$$

Factoring out a common term of $(1+r)$ isolates a standard geometric progression inside the summation block:

$$\sum_{k=1}^{f}(1+r)^k = (1+r)\sum_{k=0}^{f-1}(1+r)^k = (1+r)\cdot\frac{(1+r)^f - 1}{r} \tag{15}$$

Distributing the factored $(1+r)$ back across the numerator yields the completed, time-synchronized depletion engine expression matching equation 13.

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# 5 The Unified Derivation with Legacy Capital Floors

To complete our global solution, we enforce our final boundary constraint where our asset base at the end of life must exactly match our legacy target ($R_f = L$). By substituting our timeline identity $f = H - w$, we can map our accumulation peak directly into our depletion model.

## 5.1 The Color-Coded Substitution

We inject the accumulation boundary definition (Equation 10) directly into our synchronized depletion phase expression (Equation 15), creating a unified timeline equation:

$$\left[A_0(1+r)^w + c\cdot\frac{(1+r)^w - 1}{r}\right](1+r)^{H-w} - b\cdot\frac{(1+r)^{H-w+1} - (1+r)}{r} = L \tag{16}$$

$$\left[A_0(1+r)^w + c\cdot\frac{(1+r)^w-1}{r}\right](1+r)^{H-w} - b\cdot\frac{(1+r)^{H-w+1}-(1+r)}{r} = L \tag{17}$$

$$\left[A_0(1+r)^w + c\cdot\frac{(1+r)^w-1}{r}\right](1+r)^{H-w} - \cdot\frac{(1+r)^{H-w+1}-(1+r)}{r} = L \tag{18}$$

## 5.2 The Algebraic Isolation

Distributing the central exponential term $(1+r)^{H-w}$ across our accumulation bracket allows the localized $w$ variables to elegantly combine and drop out of the leading terms:

$$A_0(1+r)^H + \frac{c}{r}(1+r)^H - \frac{c}{r}(1+r)^{H-w} - b\cdot\frac{(1+r)^{H-w+1}-(1+r)}{r} = L \tag{19}$$

Multiplying the entire equation by our common denominator $r$ removes fractional complexity:

$$rA_0(1+r)^H + c(1+r)^H - c(1+r)^{H-w} - b(1+r)^{H-w+1} + b(1+r) = rL \tag{20}$$

Grouping the terms containing our targeted timeline variable $w$ onto the right-hand side, and moving our legacy parameter to the left-hand side:

$$(1+r)^H(rA_0+c) + b(1+r) - rL = [c + b(1+r)](1+r)^{H-w} \tag{21}$$

Isolating our exponential timeline block completely:

$$(1+r)^{H-w} = \frac{(rA_0+c)(1+r)^H + b(1+r) - rL}{c+b(1+r)} \tag{22}$$

Taking the natural logarithm (ln) of both sides frees our timeline variable from the exponent:

$$(H-w)\ln(1+r) = \ln\left[\frac{(rA_0+c)(1+r)^H + b(1+r) - rL}{c+b(1+r)}\right] \tag{23}$$

Isolating $w$ yields the final closed-form analytical solution for the required working timeline:

$$w = H - \frac{\ln\left[\frac{(rA_0+c)(1+r)^H + b(1+r) - rL}{c+b(1+r)}\right]}{\ln(1+r)} \tag{24}$$

Observe that evaluating the limit of Equation 22 as $r \to 0$ collapses perfectly into our baseline linear case derived in Equation 9, confirming absolute mathematical symmetry across all domains.

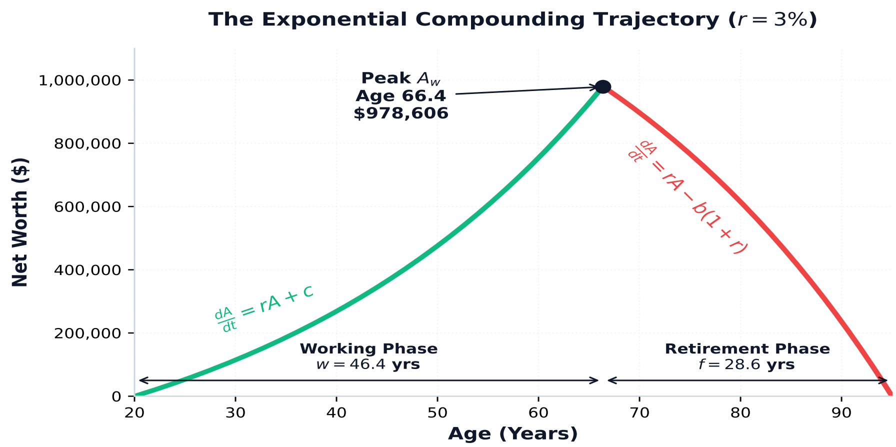


Figure 2: The exponential compounding lifecyle curves where $r > 0\%$. The accumulation phase curves upward via geometric expansion and transitions to an accelerated annuity-due depletion phase.

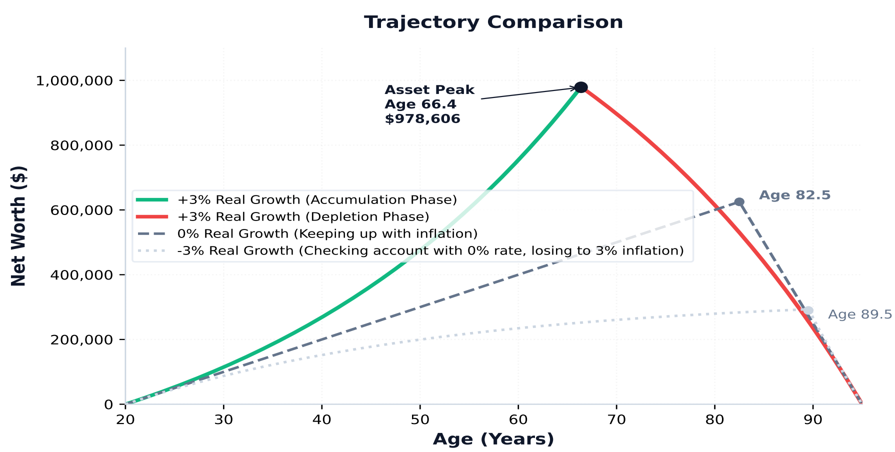


Figure 3: The exponential compounding curve compared to both assets just keeping with inflation and assets losing to inflation.